SENTINEL

Sentinel

AI-Powered FinTech Compliance & Risk Platform — Real-time AML Screening · Fraud Detection · Merchant Onboarding

TEAM
Shreyash Singh & Group

GUIDE
Dept. of CSE-AIML

INSTITUTION
IPS Academy, Indore (RGPV)

SESSION
2026–27 · B.Tech Major Project-I

SLIDE 02 — THE STATE OF COMPLIANCE

The compliance crisis in FinTech

Transaction volumes are exploding. Rule-based systems can't keep up. Regulators are demanding real-time, explainable decisions — and the cost of getting it wrong is no longer survivable.

94%

False-positive rate of legacy rule-based AML systems

SOURCE · INDUSTRY BENCHMARKS 2024

311%

Surge in synthetic identity fraud, Q1 2024 → Q1 2025

SOURCE · FEDERAL RESERVE 2024

$2.7B

Average fine per major bank for non-compliance

SOURCE · IJPREMS 2025

- UPI alone processes **15B+ payments per month** — manual review queues can't scale.
- Traditional alerts are processed in **hours, not seconds** — the real-time mandate from RBI, FATF, OFAC makes this untenable.
- Reputational damage from compliance failures is **unrecoverable** even when fines are paid.

SLIDE 03 — MARKET GAP

Why current systems fail

Three approaches dominate the market. All three lose on either openness, explainability, or unified coverage.

	LEGACY RULE ENGINES	LEGACY ML PIPELINES	COMMERCIAL REGTECH SAAS
Decision logic	✕ Hardcoded thresholds, no context	✕ Static features, no entity understanding	✕ Closed, opaque, $5K+/month
Identity coverage	✕ Misses aliasing & transliteration	✕ Misses fraud rings & collusion	✕ Vendor lock-in, no explainability
Alert quality	✕ 94% false positives	✕ Cold-start on new fraud patterns	✕ Audits become black boxes
Operations	✕ Manual alert triage	✕ No autonomous investigation	✕ No unified AML + Fraud + KYB

● Sentinel The market is **fragmented, closed, and explainability-poor**. Sentinel is open, unified, and explainable.

SLIDE 04 — WHAT WE SET OUT TO BUILD

Project objectives

Five concrete goals — each measurable, defensible, and traceable to a research basis.

01

Unified platform

AML screening, fraud detection, and merchant onboarding — not three separate tools.

COVERAGE · **3-IN-1**

02

Cut false positives ≥ 50%

Vector fuzzy matching + LLM entity resolution against exact-match sanctions screening.

TARGET · **≥ 50%**

03

Detect fraud rings via GNN

Graph Neural Networks on the Neo4j entity graph catch collusion and synthetic identity rings.

TARGET · **≥ 95% AUC-ROC**

04

Onboard in < 5 minutes

RAG-based document extraction for low-risk applicants — instead of multi-day KYB reviews.

TARGET · **DAYS → MINUTES**

05

Open & explainable

Production-grade, fully open-source alternative to closed RegTech SaaS — regulator-ready audit trail.

LICENSE · **OPEN SOURCE**

SLIDE 05 — THE PLATFORM

One platform, three pillars

A shared event-streaming backbone means risk context flows between pillars in real time — not in nightly batch jobs.

PILLAR 01

Transaction screening

Real-time AML & sanctions screening using LLM entity resolution + Milvus vector fuzzy matching.

Recall · **98%** · F1 **98.23%**

PILLAR 02

Fraud detection

Behavioral profiling + Graph Neural Network fraud-ring detection + agentic alert triage.

Accuracy · **97.5%** · vs 93.2% traditional ML

PILLAR 03

Merchant approval

KYB/KYC workflow with RAG document extraction, web scraping, and dynamic risk scoring.

Onboarding · **days → <5 min**

UNIFYING LAYER | Apache Kafka event backbone streams risk context between all three pillars in real time — every decision sees what the others see.

System architecture

A six-layer stack — each layer independently deployable, observable, and replaceable.

L1 Client layer	**Web dashboard** React SPA	**Mobile API** REST + WebSocket	**3rd-party integrations** Webhook & SDK
L2 API gateway	**FastAPI + Uvicorn** async, OpenAPI 3	**JWT auth** RBAC scopes	**Rate limiting** token-bucket
L3 Event streaming	**Apache Kafka** real-time backbone		**Redis** cache + broker
L4 Intelligence	**Screening engine** LLM + vector match	**Fraud engine** GNN + ML ensemble	**Merchant engine** RAG + risk scoring
L5 Data layer	**PostgreSQL 16** ACID relational	**Milvus** vector store	**Neo4j 5** fraud graph
L6 Observability	**Prometheus** metrics	**Grafana** dashboards	**GitHub Actions** CI/CD

Core technology stack

Battle-tested open-source components — no proprietary dependencies, no vendor lock-in.

BACKEND

FastAPI | Python 3.11 | Pydantic

STREAMING

Apache Kafka | Redis Streams

DATABASES

PostgreSQL 16 | Milvus | Neo4j 5

AI / LLM

LangChain | LlamaIndex | HuggingFace Transformers

ML MODELS

XGBoost | PyTorch Geometric | sentence-transformers

OCR

Tesseract | EasyOCR | Dual-engine routing

INFRA

Docker | Docker Compose | Kubernetes (prod) | Prometheus | Grafana | GitHub Actions

TESTING

pytest | 42 tests · 52% coverage | Locust (load)

SLIDE 08 — PILLAR 01

Intelligent transaction screening

Five-stage pipeline. Every transaction gets an entity-resolved, vector-matched, rule-and-ML-scored disposition — with full audit trail.

01 / INGEST
Transaction received
POST /api/v1/screening/transactions

›

02 / RESOLVE
NER + entity resolution
LangChain LLM — transliteration, aliases, corporate variants

›

03 / MATCH
Vector fuzzy match
Milvus cosine similarity against OFAC / EU / UN

›

04 / SCORE
AML rule engine
structuring, high-risk jurisdictions, volume anomalies

›

05 / DECIDE
Hybrid disposition
deterministic rule + probabilistic ML → final verdict

KEY CAPABILITY | **OpenSanctions-Pairs style LLM entity matching**

F1 SCORE | 98.23% vs 95.94% traditional

RESEARCH BASIS | arXiv 2411.05815

Fraud detection & agentic triage

Two layers: a detection ensemble catches the pattern; an autonomous LLM agent pre-investigates so analysts get dossiers, not raw alerts.

Layer A — Detection

GNN + BEHAVIORAL + ENSEMBLE

- Behavioral profiling — per-user / per-merchant velocity, deviation, device fingerprint
- **Graph Neural Network** on Neo4j — fraud rings, collusion, synthetic identities
- XGBoost ensemble for tabular fraud signals
- Sub-200ms inference on streaming transactions

Accuracy · **97.5%** (GNN-enhanced) vs 93.2% traditional ML

Layer B — Agentic triage

THE DIFFERENTIATOR

- Autonomous LLM agent performs **pre-investigation** of high-risk alerts
- Correlates across transactions, devices, IPs, behavioral history, graph neighborhood
- Hands a summarized **risk dossier** to the human analyst — not a raw alert

alert_id	ALT-9F3A2C
entity	user_8821@upi
graph_neighbors	14 flagged
velocity_anomaly	+840%

Likely synthetic identity ring — recommend immediate freeze on 3 linked accounts.

SLIDE 10 — PILLAR 03

Automated merchant onboarding

Five steps from application to disposition. Low-risk applicants are approved in under five minutes; risky ones get escalated with full context.

01

Application

Merchant submits docs via `POST /api/v1/merchant/onboard`

INPUT

02

Document extraction

LlamaIndex RAG pulls facts from incorporation docs, tax IDs, bank statements

RAG

03

Web intelligence

Adverse media, sanctions re-screening, reputation signals

SCRAPE

04

Risk scoring

Document confidence + behavioral priors + network signals

SCORE

05

Disposition

Auto-approve · Manual review · Auto-reject (sanctioned / fraud-linked)

AUTO / MANUAL

IMPACT | Onboarding time **days → < 5 minutes** for low-risk applicants

COMPLIANCE | Full audit trail in PostgreSQL with **deterministic explainability**

Implementation roadmap · 20 weeks, 5 phases

Each phase ships a working, testable slice of the system — not a long waterfall of partial work.



Phase 1 · Foundation

WEEKS 1 – 4

Repo, Docker Compose (10 services), Postgres schema, Milvus collections, FastAPI skeleton

Phase 2 · Screening engine

WEEKS 5 – 8

AML rule engine, entity resolution, vector fuzzy matching

Phase 3 · Fraud detection

WEEKS 9 – 12

Behavioral profiling, GNN fraud rings, agentic alert triage

Phase 4 · Merchant approval

WEEKS 13 – 16

KYB / KYC pipeline, RAG extractor, dynamic risk scoring

Phase 5 · Integration & hardening

WEEKS 17 – 20

End-to-end pipeline, CI/CD, monitoring, documentation

Results & validation

Numbers are measured against published research baselines — not cherry-picked demos.

AML SCREENING

98 %

↑ 60% reduction in false positives vs exact-match

arXiv 2411.05815 · OpenSanctions Pairs

FRAUD DETECTION

97.5 %

↑ vs 93.2% traditional ML on the same dataset

arXiv 2024 · GNN fraud detection (217 citations)

MERCHANT ONBOARDING

< 5 min

↑ days → minutes for low-risk applicants

arXiv 2501.15290 · RAG for fraud

CODE QUALITY

42 tests passing · **52% coverage** · 10-service Docker Compose deploys cleanly

REPRODUCIBILITY

All numbers traced to **arXiv 2411.05815, 2603.11051, 2501.15290** · full provenance in README

Impact, limitations & future work

Honest about what's done, what's open, and where this goes next.

IMPACT

What this changes

- **Open-source alternative** to closed RegTech SaaS (Sardine, Feedzai, Checkmarble)
- **Explainable by design** — every alert has a deterministic audit trail, regulator-ready
- **Unified platform** solves market fragmentation across AML + Fraud + KYB
- Production-grade Python/FastAPI codebase, full Docker Compose stack, CI/CD

LIMITATIONS

What's open

- Tested against synthetic + open-source datasets (OFAC, EU sanctions) — not yet on production data
- LLM calls add latency & cost — needs caching & model distillation for scale
- GNN training requires labeled fraud-ring data (cold-start challenge)

FUTURE WORK

Where this goes

- Real-time streaming inference with **Flink** + **vLLM / Triton**
- Federated learning across institutions without sharing raw data
- **Voice / chat analyst copilot** for guided investigation
- FedRAMP / SOC2 compliance pathway for US enterprise sales

Open to collaboration · Code, issues, and PRs welcome

`celestialgeeks/sentinel-fintech`